

# High-Level Design (HLD)

## ROSHN Parametric Masterplan Modelling Platform

**Version:** 1.0 **Date:** 5 January 2026 **Status:** Production Ready

**Note:** The application stack is certified production-ready. Any future enhancements or modifications based on ROSHN feedback will be implemented as iterative improvements and do not affect the current production readiness status.

### 1. Executive Summary

The ROSHN Parametric Masterplan Modelling Platform is a web-based application designed to calculate and model construction costs for real estate development projects. The platform enables cost estimation through parametric modelling, benchmark comparison, and multi-phase project planning.

### 2. System Overview

#### 2.1 Purpose

The platform provides:

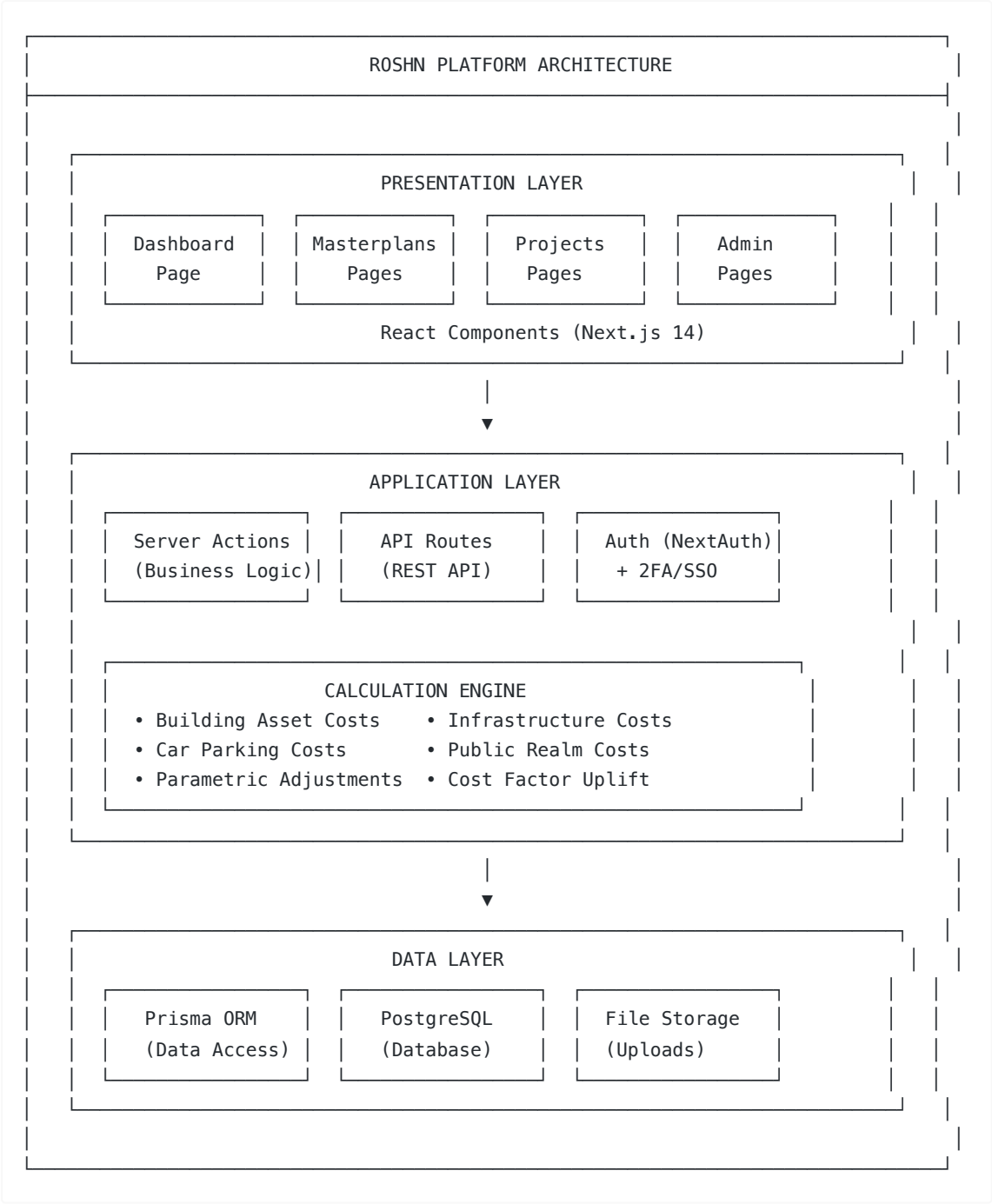
- Parametric cost modelling for masterplan developments
- Benchmark project comparison and analysis
- Multi-phase project timeline planning
- Role-based access control with team collaboration
- Comprehensive audit logging and reporting

#### 2.2 Key Features

| Feature              | Description                                                       |
|----------------------|-------------------------------------------------------------------|
| Masterplan Modelling | Create and manage development cost estimates                      |
| Cost Calculations    | Automated building, infrastructure, and parking cost calculations |
| Benchmark Projects   | Import and compare against historical project data                |
| Phase Planning       | Multi-phase timeline with quarterly scheduling                    |
| Team Collaboration   | Assign team members with role-based permissions                   |
| Configuration        | Customizable cost factors and parametric matrices                 |
| Audit Trail          | Complete activity logging for compliance                          |

### 3. Architecture Overview

#### 3.1 System Architecture Diagram



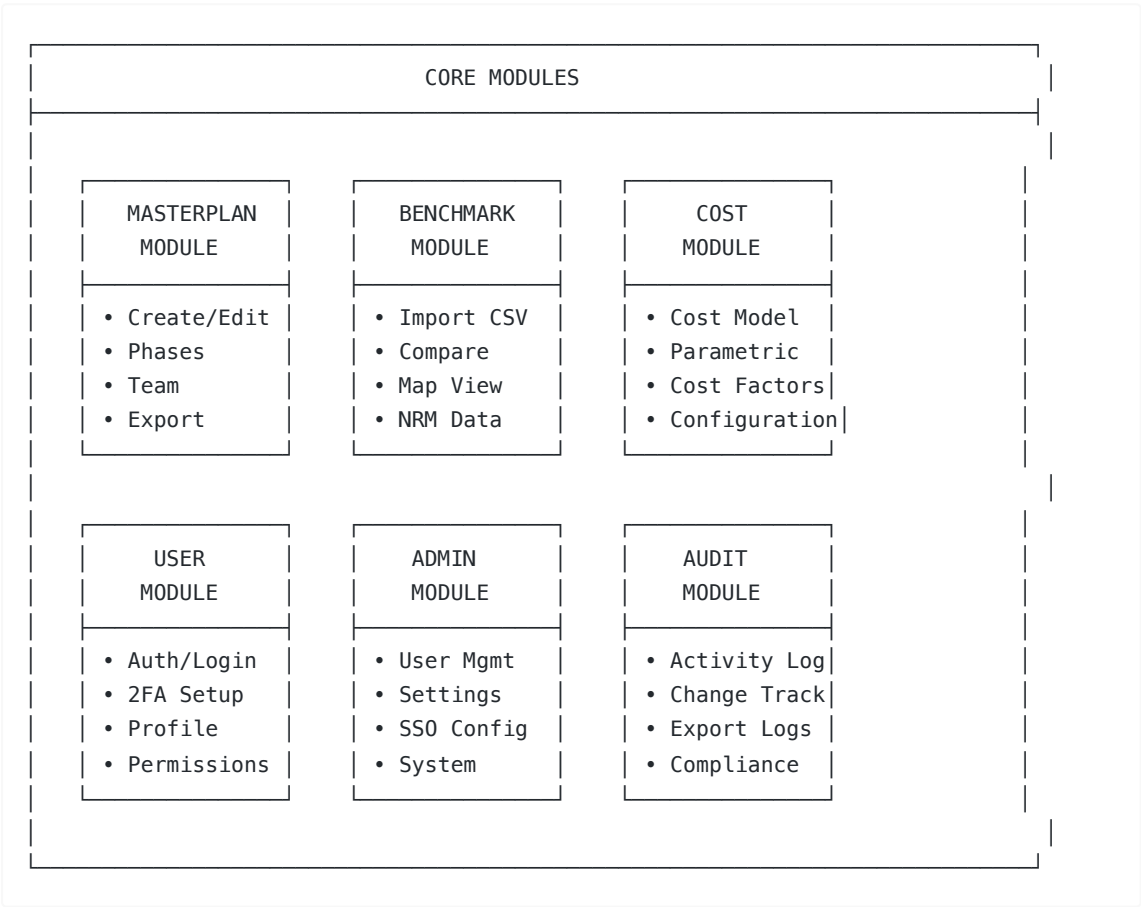
3.2 Technology Stack

| Layer    | Technology                         | Purpose                              |
|----------|------------------------------------|--------------------------------------|
| Frontend | Next.js 14, React 18, TypeScript   | User interface and client-side logic |
| Styling  | Tailwind CSS, shadcn/ui            | Component styling and UI library     |
| State    | React Query, Zustand               | Data fetching and client state       |
| Backend  | Next.js API Routes, Server Actions | Business logic and API endpoints     |

|                 |                |                                      |
|-----------------|----------------|--------------------------------------|
| Authentication  | NextAuth.js v5 | User authentication with 2FA and SSO |
| Database        | PostgreSQL 14+ | Relational data storage              |
| ORM             | Prisma         | Database access and migrations       |
| Process Manager | PM2            | Production process management        |
| Reverse Proxy   | Nginx          | Load balancing and SSL termination   |

## 4. Component Architecture

### 4.1 Core Modules



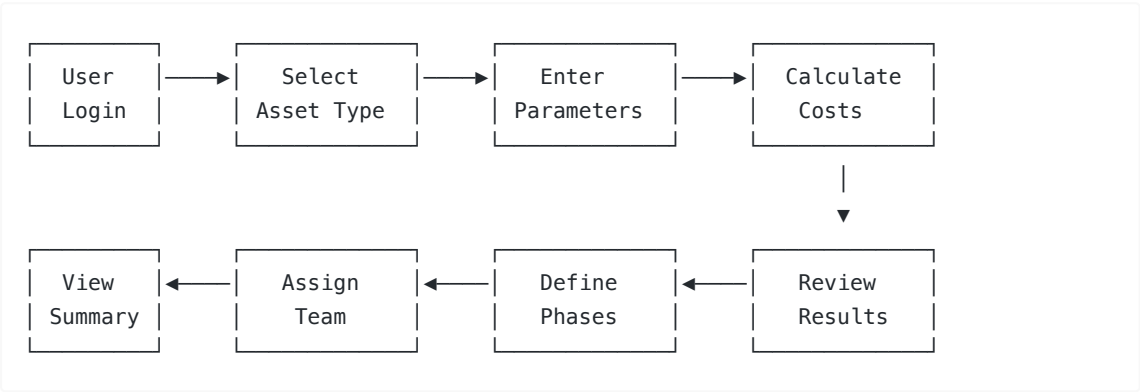
### 4.2 Module Descriptions

| Module            | Responsibility                                                        |
|-------------------|-----------------------------------------------------------------------|
| <b>Masterplan</b> | Create, edit, and manage masterplan projects with multi-phase support |
| <b>Benchmark</b>  | Import historical projects and compare costs across developments      |
| <b>Cost</b>       | Manage cost model entries, parametric matrices, and cost factors      |
| <b>User</b>       | Authentication, authorization, profile management, and 2FA            |

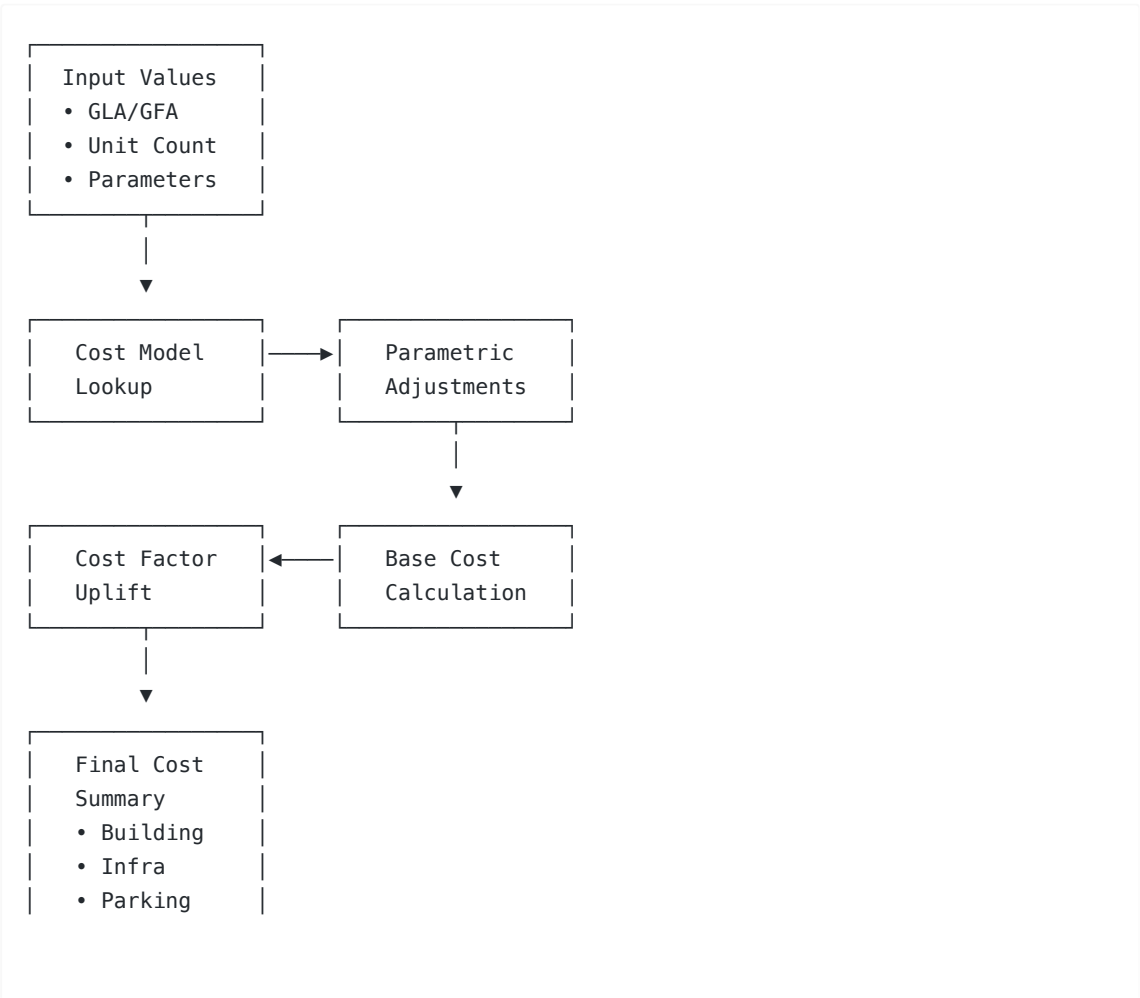
|              |                                                             |
|--------------|-------------------------------------------------------------|
| <b>Admin</b> | System administration, user management, and configuration   |
| <b>Audit</b> | Activity logging, change tracking, and compliance reporting |

# 5. Data Flow

## 5.1 Masterplan Creation Flow



## 5.2 Cost Calculation Flow



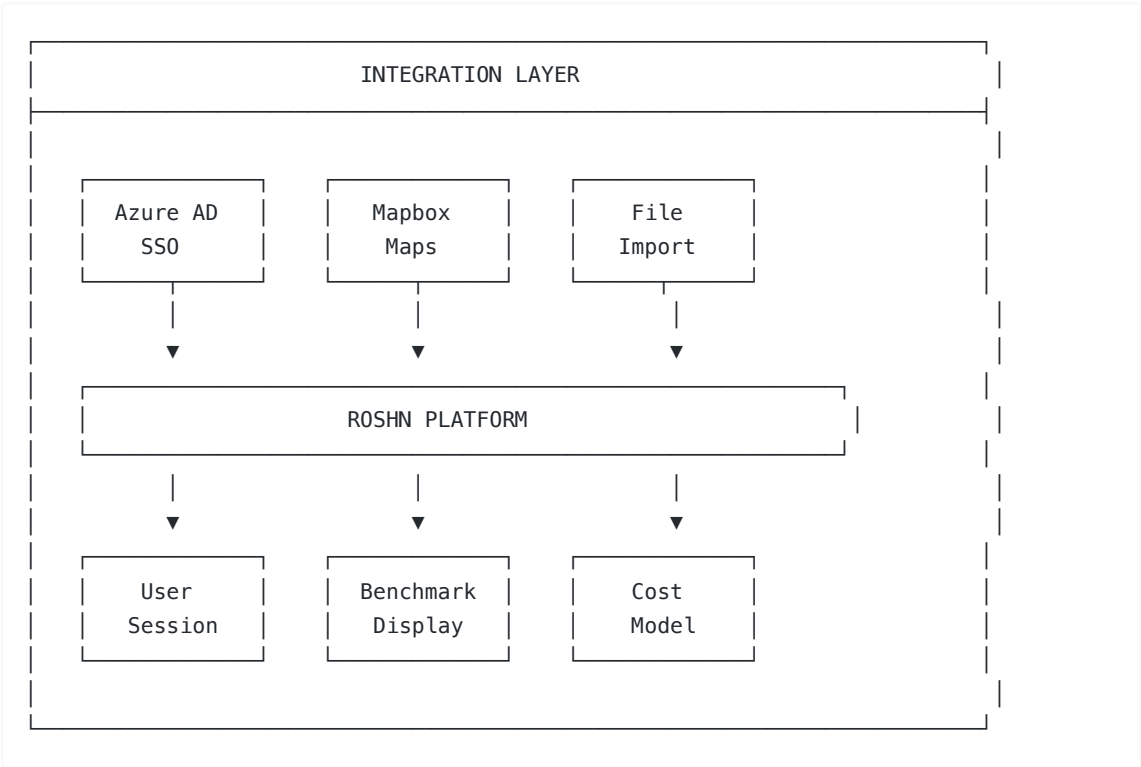
- Public Realm

## 6. Integration Points

### 6.1 External Integrations

| Integration          | Type | Purpose                                         |
|----------------------|------|-------------------------------------------------|
| Azure AD / ROSHN-IAM | SSO  | Single Sign-On authentication                   |
| Mapbox               | API  | Map visualization for benchmark projects        |
| CSV Import           | File | Bulk data import for cost models and benchmarks |
| PDF/Excel Export     | File | Report generation and data export               |

### 6.2 Integration Architecture



## 7. Security Architecture

### 7.1 Security Layers



Layer 1: Network Security

- HTTPS/TLS encryption
  - VPN access
- Firewall rules
  - Rate limiting

Layer 2: Application Security

- JWT session tokens
  - Input validation
- CSRF protection
  - SQL injection prevention

Layer 3: Authentication

- Password hashing (bcrypt)
  - Account lockout
- Two-Factor Auth (TOTP)
  - SSO integration

Layer 4: Authorization

- Role-based access (RBAC)
  - Country/Developer filtering
- Project-level permissions
  - Team member roles

Layer 5: Data Security

- Encrypted secrets
  - Audit logging
- Database encryption
  - Data backup

7.2 Role-Based Access Control

| Role                | Permissions                                        |
|---------------------|----------------------------------------------------|
| ADMIN               | Full system access, user management, configuration |
| DEVELOPMENT_MANAGER | Create/edit masterplans, manage assigned projects  |
| VIEWER              | Read-only access to assigned projects              |

7.3 Team Roles (Project Level)

| Role    | Permissions                                           |
|---------|-------------------------------------------------------|
| MANAGER | Full access to specific project, can edit masterplans |
| VIEWER  | Read-only access to specific project                  |

8. Scalability & Performance

8.1 Scalability Strategy

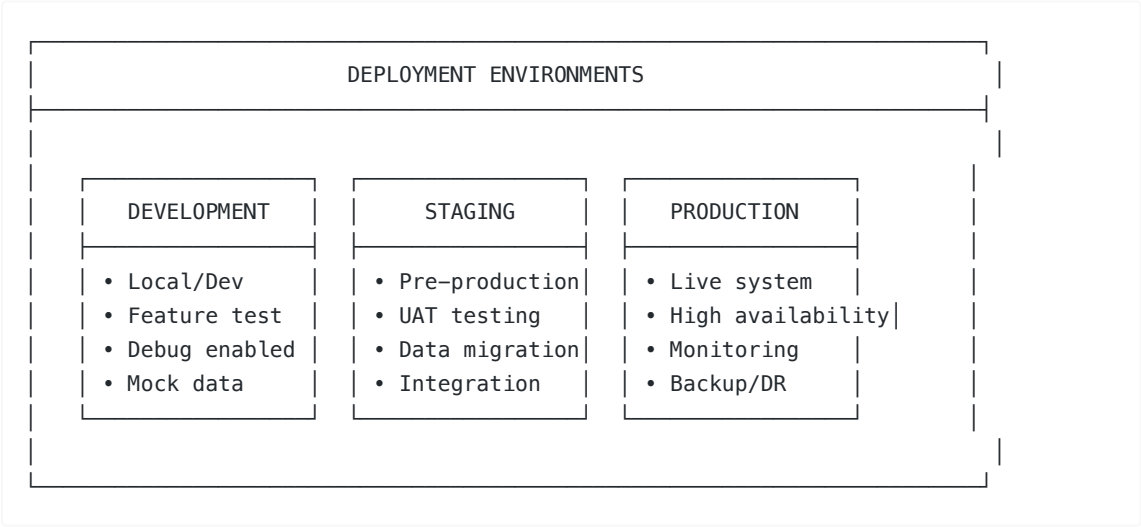
| Aspect             | Strategy                                         |
|--------------------|--------------------------------------------------|
| Horizontal Scaling | PM2 cluster mode, Cloud Run auto-scaling         |
| Database           | Connection pooling, read replicas for reporting  |
| Caching            | React Query caching, database query optimization |
| CDN                | Static asset delivery via CDN                    |

8.2 Performance Targets

| Metric              | Target            |
|---------------------|-------------------|
| Page Load Time      | < 2 seconds       |
| API Response Time   | < 500ms           |
| Database Query Time | < 100ms           |
| Concurrent Users    | 100+ simultaneous |

9. Deployment Architecture

9.1 Environment Overview



9.2 Deployment Options

| Platform           | Use Case                    | Documentation                    |
|--------------------|-----------------------------|----------------------------------|
| Microsoft Azure    | UAE-based cloud hosting     | <a href="#">Azure Guide</a>      |
| Google Cloud (GCP) | Saudi Arabia data residency | <a href="#">GCP Guide</a>        |
| On-Premise         | Full data sovereignty       | <a href="#">On-Premise Guide</a> |

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## 10. Monitoring & Logging

### 10.1 Monitoring Strategy

| Component      | Monitoring                         |
|----------------|------------------------------------|
| Application    | PM2 metrics, health checks         |
| Database       | Connection pool, query performance |
| Infrastructure | CPU, memory, disk usage            |
| Security       | Failed logins, suspicious activity |

### 10.2 Logging Strategy

| Log Type         | Retention | Purpose                       |
|------------------|-----------|-------------------------------|
| Application Logs | 30 days   | Debugging and troubleshooting |
| Audit Logs       | 1 year    | Compliance and security       |
| Access Logs      | 90 days   | Security analysis             |
| Error Logs       | 30 days   | Issue resolution              |

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## 11. Disaster Recovery

### 11.1 Backup Strategy

| Data Type     | Frequency | Retention |
|---------------|-----------|-----------|
| Database      | Daily     | 30 days   |
| Configuration | On change | 90 days   |
| File Uploads  | Daily     | 30 days   |

### 11.2 Recovery Objectives

| Metric                         | Target   |
|--------------------------------|----------|
| Recovery Time Objective (RTO)  | 4 hours  |
| Recovery Point Objective (RPO) | 24 hours |

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## 12. Compliance & Standards

### 12.1 Compliance Requirements

| Standard | Status |
|----------|--------|
|----------|--------|



|                   |             |
|-------------------|-------------|
| Data Privacy      | Compliant   |
| Audit Trail       | Implemented |
| Role-Based Access | Implemented |
| Encryption (TLS)  | Enabled     |
| Password Policy   | Enforced    |

**12.2 Security Standards**

- OWASP Top 10 protection
- Input validation and sanitization
- SQL injection prevention (Prisma ORM)
- XSS protection (React sanitization)
- CSRF protection (NextAuth)